

## Lesson 1.16: Unit Price and Comparison Shopping

**Unit:** Unit 1, Kitchen Skills and Smart Eating (Nutrition and Wellness) **Topic:** 1.4 Plan, Shop, Cook: Meals on a Budget **Grade:** 6 to 8 (written at grade 7; see Differentiation for the grade 6 support and grade 8 stretch) **Days:** 1 class period of 40 minutes **Room:** Either (classroom or FACS lab; no cooking today)

### Standards

- NYS module line: NYS N&W 3. Consumer Resources and Finance b) Describe and demonstrate food purchasing strategies that maximize the benefits of the food budget
- NYS module line: NYS N&W 3. Consumer Resources and Finance c) Identify monetary influences on one's ability to select and purchase nutritious foods
- CTE theme line: CTE FCL 4. Buying Goods and Services f) Apply a decision-making model to maximize consumer satisfaction when purchasing goods and services
- CTE theme line: CTE FCL 4. Buying Goods and Services b) Distinguish between needs, wants, values, and goals and tell how each impacts spending and savings decisions
- NYS FACS Learning Standard 3 (Intermediate), Key Idea: Students will understand and be able to manage personal resources of talent, time, energy, and money and make effective decisions in order to balance their obligations to work, family, and self.; Performance Indicator: Students understand the resources available to them, make informed decisions about the use of those resources, and know some ways to expand resources.
- CDOS Standard 3a Universal Foundation Skills: Basic Skills
- National FCS 3.0: 2.1.3 Examine the decisions behind providing food that is safe and nourishing for individuals and families.
- National FCS 3.0: 2.5.1 Examine how resources get used when individuals, families, and communities choose among needs and wants.

(Note: the unit plan suggested 3.3.4. Its 3.0 text is "Compare investment and savings alternatives," which is about savings accounts, not groceries, so it is not cited. 2.1.3 and 2.5.1 are the confirmed 3.0 lines that fit unit price and comparison shopping.)

### Enduring understanding and essential question

**Big idea:** The price on the front of the package tells you what you pay. The unit price tells you what you get for it. The unit price is the only fair way to compare two sizes, two brands, or two stores. **Essential question:** When is the cheaper thing not the better buy?

### Objectives

By the end of this lesson, students will be able to:

1. (Low) Given a price and a quantity, calculate the unit price (price divided by quantity) to the nearest cent.

2. (Mid) Given two or three versions of the same item (store brand and name brand, small and bulk), compare their unit prices and select the better buy.
3. (High) Given a scenario where the lowest unit price is the wrong choice (too much to use before it spoils, no storage space, a family rule), evaluate the choice and explain in two sentences why “cheaper per ounce” lost.

## Vocabulary

Tier 2 (general academic): compare, quantity, per, ratio Tier 3 (FACS): unit price, shelf tag, store brand (private label), name brand, bulk, sales tax (on non-food items), perishable. See the unit vocabulary list for the Spanish column.

## Materials and setup

- Handout, one per student: Handout 01.16 - Unit Price Practice.md
- The circulars from Lesson 1.15, one per pair
- Real packages or photos of three item pairs for the demo: a 16 oz store-brand pasta box and a 12 oz name-brand box; a 24 oz jar of sauce and a 45 oz jar; an 8 oz bag of shredded cheese and a 2 lb bag
- Calculators, one per pair (phones are fine if the room allows)
- Slides: Slides 01.16 - Unit Price and Comparison Shopping (outline in 05 Materials; the site renders it as a deck) (includes a photo of a real shelf tag with the unit price called out)
- Setup notes: Do the demo math yourself first so the numbers on the slides match the packages in your hand. Allergy check: sealed packages only; nothing is opened.

## Pre-assessment

The do now is a straight division problem. Anyone who gets it in under a minute can be a peer checker in Step 3. Anyone who cannot needs the calculator and the worked example on the handout, not a speech about division.

## Do now and hook (Time: 5 min)

On the board: “A 16 ounce box of pasta costs \$1.29. A 12 ounce box of a different brand costs \$1.49. Which is the better buy? Show your thinking.”

Debrief: most students will pick the \$1.29 box because the number is smaller. Ask: “Is it smaller because it is a better deal, or because there is less in the box?” There is more in the box and it costs less, so this one is easy. Then the hook: hold up two jars of sauce, 24 oz at \$2.29 and 45 oz at \$3.79. “This time the big one costs more. Is it the better buy? You cannot tell by looking. By the end of today you will be able to tell in ten seconds.”

## Procedure

| Step                                      | Teacher will                               | Students will                   | Time  |
|-------------------------------------------|--------------------------------------------|---------------------------------|-------|
| 1. Direct instruction: unit price and the | Write the rule: unit price = price divided | Copy the rule and the two sauce | 7 min |

|                                             |                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                              |        |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|--------|
| shelf tag                                   | by quantity. Do the two sauce jars on the board: \$2.29 divided by 24 = \$0.095 per oz; \$3.79 divided by 45 = \$0.084 per oz. Show a real shelf tag photo and point to the small unit price the store already prints. Rule two: the unit has to match (oz to oz, lb to lb) before you compare. Show 1 lb = 16 oz.                                                                                                        | answers on the handout. Answer: "Which jar wins per ounce?"                                                                  |        |
| 2. Direct instruction: brand, bulk, and tax | Three quick comparisons with real packages: store brand vs name brand (usually same factory, different label, 20 to 40 percent less); small vs bulk (bulk usually wins per ounce, but only if you can use it and store it); a food item vs a non-food item at the register (in New York most grocery food is not taxed; paper plates, foil, cups, soap, and candy or soda are taxed, about 8.625 percent on Long Island). | Predict the winner before each reveal with a thumb up for store brand or bulk. Write the tax rule in the box on the handout. | 6 min  |
| 3. Practice: Unit Price Practice sheet      | Pairs work through the handout: six unit price calculations from circular items, three "which is the better buy" pairs, and one bulk trap (a                                                                                                                                                                                                                                                                              | Calculate, compare, circle the better buy, and write the bulk trap answer.                                                   | 12 min |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                    |                                                |       |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|-------|
|                                        | 10 lb bag of potatoes for a family that eats potatoes once a month). Circulate; check the unit conversions first. Peer checkers from the do now help pairs that are stuck.                                                                                                                                                                                                         |                                                |       |
| 4. Debrief: when cheaper is not better | Take answers to the bulk trap. Build the list on the board: cheaper per unit is the wrong buy when it will spoil first, when there is no room to store it, when the family will not eat that much, when a food rule or allergy rules it out, or when you cannot afford the bigger total today even though it is cheaper per ounce. That last one is the hardest and the most real. | Copy the list. Add one of their own from home. | 6 min |

### Questions to ask

Monitor understanding:

- \$4.99 for a 2 lb bag. What is the price per ounce? What did you do first?
- The shelf tag says the store brand is \$0.12 per oz and the name brand is \$0.19 per oz. Same food?
- Which of these gets taxed at the register: a bag of rice, a roll of foil, a bottle of soda?

Deepen learning:

- Why would a store print the unit price in tiny letters and the sale price in huge ones?
- A family with \$40 for the whole week cannot buy the \$12 bulk chicken even though it is cheapest per pound. What would you tell them to do instead?
- Is there anything your family always buys name brand? Why? Is that a want, a need, or a value?

## Check for understanding

Whiteboard check at the end of Step 1: put a price and a quantity on the screen; every pair writes the unit price and holds it up. A “got it” answer is within a cent and has the right unit written. Then clipboard during Step 3: a yes or no per pair on “converts pounds to ounces before comparing.” A no gets a question at the pair’s table, not a class stop.

## Closure (Time: 4 min)

Exit card: 3 steps to find a unit price (find the price, find the quantity in the same unit, divide), 2 reasons the cheaper unit price can be the wrong buy, 1 item your family buys where you now want to check the shelf tag. Three students read one line aloud. Sort into “got it” and “needs more”; anyone in “needs more” on the math gets the worked-example card taped into their project folder before Lesson 1.18.

## Differentiation and supports

- ELL: the rule is on the slide as a formula and as a picture (a dollar sign over a package); the shelf tag photo has the unit price circled; sentence starter for the bulk trap: “The big bag is cheaper per , **but this family should buy the small one because**”; the words price, unit, ounce, pound, brand, and tax are on the unit vocabulary card in Spanish.
- IEP and 504 (general): calculators for everyone; the handout gives the quantity already in ounces for the first four problems so the conversion is separate from the division; a worked example at the top of every section; extended time; the bulk trap can be answered aloud to the teacher.
- Grade 6 support: all quantities are in the same unit (no pound-to-ounce conversion); the practice sheet is four calculations and two comparisons.
- Grade 8 stretch: add the pound-to-ounce conversions on every problem, compare the same item across two circulars, and calculate the sales tax on the household items in the Lesson 1.20 fallback station cards to the penny.
- UDL checkpoint used: multiple means of representation (the rule as words, as a formula, and as a real shelf tag) and multiple means of action and expression (whiteboard, handout, or spoken answer).
- No-kitchen or no-machine alternative: not needed; paper and calculators. If no real packages are available, the slides carry photos with the prices printed.

## Assessment

| Objective                 | Standards met                   | Learning task          | Assessment (formative or summative; tool)                                                          |
|---------------------------|---------------------------------|------------------------|----------------------------------------------------------------------------------------------------|
| 1. Calculate a unit price | N&W 3. b); CDOS 3a Basic Skills | Practice sheet, Step 3 | Formative; six calculations checked against the teacher key, five of six within a cent is “got it” |

|                                                                  |                                                      |                                      |                                                                                                |
|------------------------------------------------------------------|------------------------------------------------------|--------------------------------------|------------------------------------------------------------------------------------------------|
| 2. Compare and select the better buy                             | N&W 3. b); FCL 4. f); Std 3 PI B; National FCS 2.1.3 | Better buy pairs, Step 3             | Formative; three comparisons checked; handout collected as daily work                          |
| 3. Evaluate a case where the cheaper unit price is the wrong buy | N&W 3. c); FCL 4. b); National FCS 2.5.1             | Bulk trap and debrief, Steps 3 and 4 | Formative; two sentences scored on whether the reason is one of the five from the debrief list |

Objectives 1 and 2 are assessed summatively on the Unit 1 test (the unit price math item) and through the project plan (Rubric 01, criterion 1, where every price on the plan must carry its unit).

### Homework

None. Optional: photograph one shelf tag at a store this week (with permission) and bring the photo or write down the unit price. Two minutes at the start of Lesson 1.17 for anyone who did.

### Connections

Math: this is ratio and unit rate, which is in the grade 6 and 7 math standards; the math teacher gets a real-world set of unit rate problems, and FACS gets students who have already seen the word “per.” The same circular works in both rooms the same week.

### Sources

- NYS Middle Level CTE FACS Content Module 05, Nutrition and Wellness (nyctecenter.org, June 2018), lines 3. b) and 3. c). Copy in 02 Standards/NYS Module PDFs/.
- NYS Middle Level CTE Theme Module 03, Financial and Consumer Literacy, lines 4. b) and 4. f). Same folder.
- NYS Learning Standards for Health, Physical Education, and Family and Consumer Sciences (1996), Standard 3, Intermediate. Wording per 02 Standards/STANDARDS CROSSWALK.md, Part A.6.
- CDOS Intermediate Standard 3a, per 02 Standards/STANDARDS CROSSWALK.md, Part A.7.
- National Standards for Family and Consumer Sciences Education 3.0 (LEAD FCS Education, 2018), competencies 2.1.3 and 2.5.1, per 02 Standards/STANDARDS SOURCES - web research.md.
- New York State sales tax treatment of food: most food for home use is exempt; prepared food, candy, soda, and non-food items are taxable.

- My own materials: the supermarket shopping list with store-brand swaps, the “NYS sales tax note” in my three cooking lesson packages, and the unit price day in my draft unit calendar.

### Teacher notes

- If time runs short, cut Step 2 to store brand vs name brand only and move the tax rule to Lesson 1.18, where the foil and plates on the station card make it real.
- Common mistake: dividing quantity by price instead of price by quantity. The answer comes out as “ounces per dollar,” which is a real number and looks fine. Have students write the dollar sign on every unit price answer; if there is no dollar sign, they divided the wrong way.
- Second common mistake: comparing a per-pound price to a per-ounce price. The conversion box on the handout is there for this.
- The bulk trap is the point of the lesson for the \$40 project. A team that buys the 5 lb bag of rice “because it is cheapest” has just spent a fifth of the budget on rice.